The Rally, Not the Rule: Semiconductor Stocks at the US Export Controls and DeepSeek under Automated Natural-Experiment Validation

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analysis pass of record: 2026-07-18, `natex` v0.2.0, seed 0 (all three survey runs; kink estimator deterministic)

Abstract

We point `natex` — an automated natural-experiment discovery and validation toolkit — at weekly semiconductor stock prices around the two BIS export-control rounds (2022-10-07, 2023-10-17) and the DeepSeek crash week (2025-01-27). None of its headline "credible" verdicts survives inspection. Declared regression kinks in calendar time are "credible" at all three cutoffs (e.g. DeepSeek: $\hat{\tau} = +0.48$ log/yr, Holm $p = 1.4 \times 10^{-8}$), but each is an artifact: the DeepSeek kink is the 2025 AI rally sitting on both sides of a one-week crash — a *positive* slope change at a *negative* event — the 2022 kink collapses monotonically in bandwidth ($+0.83 \to +0.13 \to +0.05$ log/yr) with ChatGPT's release inside every bandwidth, and a "credible" 2022 RDD merely rediscovered that prices above a level lie after 2022. What survives is descriptive or null, stated as such: NVDA fell 11.6 log-points *relative to* the SOXX semiconductor index in the DeepSeek week — a level break with locally unchanged relative slope ($+0.52 \to +0.39$ log/yr), not a kink — and a difference-in-differences scan across tickers finds *no* break at the export-control dates (scan $p = 0.70$; best-fitting break 2025.95, not 2022.76), an informative null consistent with export-control costs being priced jointly with the contemporaneous AI-demand shock. On financial series, automated validation refuses honestly in places but is systematically fooled by trend regimes whenever a kink in calendar time is read causally.

1 Introduction

On 7 October 2022 the US Bureau of Industry and Security (BIS) imposed broad export controls on advanced AI accelerators bound for China, read at the time as a landmark attempt to restrict China's frontier compute [1]; on 17 October 2023 a second round closed the A800/H800 loophole. Export controls impose costs on the controlling country's own firms [3]: entity-list evidence puts US suppliers' collateral losses near \$130 billion in market value, plus lost customers, lending, and employment [5]. On 27 January 2025 a demand-side shock arrived from the opposite direction: the release of DeepSeek's R1 model triggered the largest one-day market-capitalization loss in history for Nvidia, and event studies document significantly negative abnormal returns across US semiconductor stocks [7].

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The standard tool here is the event study [10]: short windows, abnormal returns against a market model, an efficient-markets identification argument. This paper deliberately uses the wrong tool — quasi-experimental designs with *calendar time* as the running or assignment variable — and asks what an automated toolkit's validation layer does about it. The dangers are known in principle: regression discontinuity in time lacks the cross-sectional no-manipulation logic of true RDDs and is biased by ignored time-series structure [8], and a regression kink [4, 2] estimated on a calendar-time running variable is, mechanically, a before/after slope contrast. The open question is empirical: does a modern validation battery — placebo covariates and cutoffs [6], density tests, bandwidth sensitivity, scan-based localization [9] — actually catch these failure modes on real financial series? We run `natex` [11] over three declared cutoffs and report every verdict, including the refusals. The answer is asymmetric: every "credible" verdict fails inspection, while the refusals, failures, and nulls are the informative outputs.

2 Data

Weekly adjusted-close prices for NVDA, AMD, ASML, TSM, and the SOXX semiconductor ETF, 784 weeks from July 2011 to July 2026 (Yahoo Finance weeklies; not committed). The analysis variable is the log adjusted close, the running variable calendar time in years; the pooled five-ticker panel gives the kink family $n_{\text{used}} = 1960$. Descriptive contrasts use the NVDA-minus-SOXX relative log price, netting out the sector factor. The three declared cutoffs are $t = 2022.7644$ (BIS round one, 2022-10-07), $t = 2023.7918$ (BIS round two, 2023-10-17), and $t = 2025.0712$ (the DeepSeek crash week beginning 2025-01-27). One confound matters throughout: ChatGPT's release (2022-11-30) lands eight weeks *after* the first BIS cutoff, inside every bandwidth the 2022 analyses select.

3 Design and Methods

Three `natex` survey runs (v0.2.0, seed 0) supply the estimates: a pooled five-ticker run declaring the DeepSeek cutoff, a pooled run declaring the BIS-2022 cutoff, and an NVDA-only run declaring the BIS-2023 cutoff. Each survey attempts seven design families on the same panel: a declared regression kink in time (local-linear RKD [4] in the difference-in-kinks tradition of [2], cross-validated bandwidth, Holm-corrected placebo battery in the spirit of [6]), a discovered RDD (candidate role assignment over {treatment, forcing, outcome} columns, scan localization [9], McCrary-style density test, covariate placebos), a sector-by-time DiD scan (SuDDDS, permutation-calibrated), synthetic control, discovered-experiments estimation (dee), IV, and bunching.

Honest-inference notes, stated as run. These caveats are part of the record, not afterthoughts. (i) `natex`'s own kink report carries the warning that "a kink in a calendar-time running variable is a before/after slope contrast" — the causal reading is on the analyst, and this paper's point is what happens when it is taken anyway. (ii) The DeepSeek-week level break below is descriptive: no placebo distribution was run, so it carries no standard error. (iii) The DeepSeek-run DiD *failed* rather than returning a number: the intake ignored the declared numeric time column and picked the string `date` column (error of record: "`time column must be numeric: date`"). (iv) The NVDA-only synthetic control *failed* by construction — "`treated unit has no finite outcome before t0`" — a single-ticker panel has no donors. (v) The dee family returned nulls with zero usable discovered experiments (fewer than its minimum of three); IV and bunching declined as `needs_input` throughout; and where a placebo battery was vacuous we say so and treat the estimate as unvalidated.

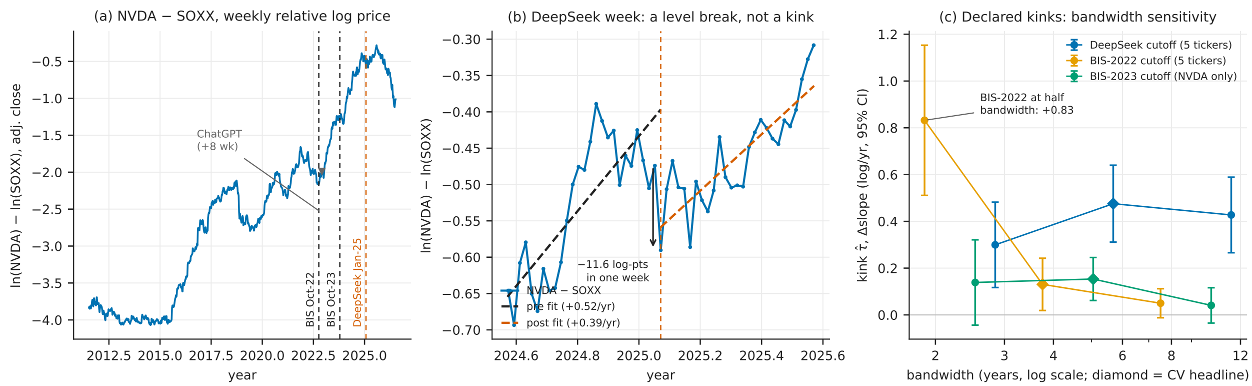

Figure 1: (a) Weekly NVDA-minus-SOXX relative log (adjusted-close) price with the three declared cutoffs; ChatGPT lands eight weeks after the BIS-2022 cutoff, and the 2023–2025 AI rally dominates the series. (b) The DeepSeek week: an 11.6 log-point one-week relative level break with locally unchanged ±26-week slope — a level break, not a kink. (c) Declared-kink estimates (95% CIs) at half, headline (diamond, cross-validated), and double bandwidth: BIS-2022 collapses monotonically, BIS-2023 fades to $t \approx 1.06$, and the DeepSeek "kink" is positive at a crash — the rally on both sides.

4 Results

The real event: a relative level break at DeepSeek. Recomputed exactly from the weekly closes, the week beginning 2025-01-27 took NVDA down −0.17211 in log terms (−17.2 log-points; −15.8% simple return) against −0.05577 for SOXX (−5.6 log-points; −5.4%): an −11.63 log-point one-week fall in NVDA *relative to its own sector*. The local relative trend barely moved: ±26-week OLS slopes of the relative log price are +0.52 log/yr before and +0.39 log/yr after (Figure 1b). The DeepSeek shock is a *level* break with an approximately unchanged local slope — the signature of a one-off repricing, not a growth-regime change. This estimate is descriptive (no placebo distribution; no standard error).

Declared kinks: "credible" three for three, wrong three for three. Table 1, panel A. At the DeepSeek cutoff the pooled kink is $\hat{\tau} = +0.4756$ log/yr (se 0.0839, 95% CI $[0.3111, 0.6400]$, Holm $p = 1.4 \times 10^{-8}$, cross-validated bandwidth 6.05 yr, $n_{\text{used}} = 1960$). The sign alone refutes the causal reading: DeepSeek was a crash, yet the "effect" is a large *positive* slope change — a six-year bandwidth spans the pre-2023 flat regime on the left and the AI rally on the right, so the kink dates the rally, not the event. At the BIS-2022 cutoff the pooled kink (+0.1305, Holm $p = 0.022$) is monotonically bandwidth-sensitive — +0.832 (se 0.164) at 1.88 yr, +0.131 (se 0.057) at 3.75 yr, +0.050 (se 0.032) at 7.50 yr — and ChatGPT's release sits eight weeks post-cutoff inside every one of those windows; nothing localizes the bend to the export controls. The NVDA-only BIS-2023 kink (+0.1534, Holm $p = 0.0011$) fades to +0.0407 (se 0.0385, $t \approx 1.06$) at double bandwidth. The ±26-week local slopes tell the same story in descriptive form: the relative NVDA-minus-SOXX slope change is +1.46 log/yr at BIS-2022 (from −0.46 to +1.00), +0.41 at BIS-2023 (from +0.61 to +1.02), and −0.13 (≈ flat, $+0.52 \to +0.39$) at DeepSeek: the two "export-control kinks" are the AI rally starting, and the one week with a genuine event shows no kink at all.

A "credible" RDD that is a role-assignment artifact. In the BIS-2022 run the discovered-RDD family also returned "credible" (scan $p = 0.010$, density $p = 0.180$). Inspection of the

Table 1: All verdicts across the three survey runs (`natex` v0.2.0, seed 0).

Run (cutoff)	family	$\hat{\tau}$	se	inference	disposition
Panel A: declared kinks in calendar time, log price (log/yr) — all "credible", none survives					
BIS-2022 (pooled)	kink	+0.1305	0.0570	Holm p=0.022; bw 3.75	+0.83 → +0.13 → +0.05 across bw
BIS-2023 (NVDA)	kink	+0.1534	0.0469	Holm p=0.0011; bw 4.77	fades: t≈1.06 at 2×bw
DeepSeek (pooled)	kink	+0.4756	0.0839	Holm p=1.4×10^{-8}; bw 6.05	positive "kink" at a crash
Panel B: every other family disposition					
BIS-2022	rdd	1.581	0.886	scan p=0.010; CI spans 0	role-assignment artifact
BIS-2022	did	—	—	scan p=0.70; best t_0=2025.95	**informative null**
BIS-2022	dee	—	—	0 < 3 usable experiments	null
BIS-2023	rdd	—	—	scan p=0.040; placebos vacuous	unvalidated
BIS-2023	did	—	—	scan p=0.76; best t_0=2017.16	null
BIS-2023	sc	—	—	no donors (single ticker)	failed, loudly
BIS-2023	dee	—	—	0 < 3 usable experiments	null
DeepSeek	rdd	4.07	2.64	scan p=1.00; CI spans 0	null (placebos worked)
DeepSeek	did	—	—	intake chose string `date` col.	failed, loudly
DeepSeek	iv/sc/bunching	—	—	`needs_input`	refused

Notes: Kink rows: local-linear RKD in calendar time, log adjusted close, cross-validated bandwidth in years, $n_{\text{used}} = 1960$; BIS-2022 disposition: estimate at half/headline/double bandwidth. 95% CIs: BIS-2022 kink $[0.0188, 0.2423]$; BIS-2023 kink $[0.0616, 0.2453]$; DeepSeek kink $[0.3111, 0.6400]$. RDD rows: fuzzy 2SLS at the discovered candidate; CIs BIS-2022 $[-0.156, 3.317]$ and DeepSeek $[-1.11, 9.25]$; density p: BIS-2022 0.180, BIS-2023 0.312, DeepSeek 0.752 with all 8 covariate placebos at Holm $p = 1.0$. DiD rows: SuDDDS permutation scan; point estimates at the (non-rejected) BIS-2022 best t_0: $\hat{\tau}_{\text{DD}} = -2.061$ (se 0.018), $\hat{\tau}_{\text{synth}} = -0.427$ (se 0.025).

selected candidate shows why this is empty: the automated role search assigned treatment = `post_2022_controls`, forcing = `log_adjclose`, and outcome = t — it mechanically rediscovered that prices above a level tend to lie after 2022. The fuzzy estimate $\hat{\tau}_{\text{2SLS}} = 1.581$ (se 0.886, 95% CI $[-0.156, 3.317]$) crosses zero and has no economic content. In the BIS-2023 NVDA-only run the RDD scan rejected at $p = 0.040$ but the covariate-placebo battery was *vacuous* (no testable covariate in the single-ticker numeric panel; density $p = 0.312$): unvalidated, and not reported as a finding.

The informative null: no DiD break at the export controls. The SuDDDS scan over tickers and break dates in the BIS-2022 run finds nothing at the controls: scan $p = 0.70$ (LLR 2.548), with the best-fitting break at $t_0 = 2025.95$ (window 3.76 yr) — not 2022.76 (point estimates at that spurious t_0 in the table notes; scan-level p-values are not computed for a non-rejected scan). The NVDA-only run's DiD is likewise null ($p = 0.76$, best $t_0 = 2017.16$). No ticker-relative break is detectable at either export-control date. This is the substantive result of the paper, and it is a null: whatever costs the controls imposed on these five firms were priced jointly with — and are not separable in these data from — the contemporaneous AI-demand shock lifting the same stocks.

The validation layer's scorecard. Table 1, panel B. Credit: at the DeepSeek cutoff the discovered-RDD family correctly returned *null* (scan $p = 1.00$, density $p = 0.752$, all eight covariate placebos at Holm $p = 1.0$; the fuzzy $\hat{\tau}_{\text{2SLS}} = 4.07$, se 2.64, was discarded), and the pipeline refused rather than fabricated where inputs were missing (`needs_input`; dee nulls; two loud failures quoted in Section 3). Debit: every "credible" verdict on these series — three kinks and one RDD — fails inspection, and cross-validated bandwidth selection actively favors the artifact, choosing windows wide enough to straddle regimes.

5 Caveats and Conclusion

Five caveats bound the claims. First, a unit correction to the working notes of record: the DeepSeek-week figures −17.2% / −5.6% are one-week *log* returns (log-points); the simple returns are −15.8% and −5.4%. Second, the level break and the ±26-week slopes are descriptive — a check on magnitudes against the formal event-study machinery [10, 7], not a substitute for it. Third, all causal-design caveats for time-as-running-variable apply with full force [8], and `natex`'s own recorded warning — a kink in a calendar-time running variable is a before/after slope contrast — should be read as governing every panel-A row. Fourth, the event window is confounded by construction: ChatGPT lands eight weeks after the BIS-2022 cutoff, and the 2023–2025 AI rally overlaps everything after it; the DiD null is therefore a statement about non-separability in these data, not evidence that the controls were costless. Fifth, five tickers are a narrow panel; the SuDDDS permutation floor and the vacuous placebo batteries both reflect that thinness.

The conclusion is a methods verdict with one substantive corollary. Methods: on trending financial series, an automated validation layer is asymmetric — its refusals, failures, and nulls were all correct and informative, while every one of its "credible" verdicts was an artifact of trend regimes, role search, or vacuous batteries; cross-validated bandwidths made the kink artifacts *more* confident, not less; "credible" on a calendar-time design is a prompt for the checks in Figure 1, not a result. Substance: the one clean event in these data is DeepSeek — an 11.6 log-point one-week repricing of NVDA against its own sector with no local trend change — while the export-control rounds left no detectable ticker-relative break at all, consistent with their costs having been priced against the contemporaneous AI-demand shock rather than as standalone news.

Reproducibility. All estimates come from three `natex` v0.2.0 survey runs at seed 0 (2026-07-18; the kink estimator is deterministic) on frozen weekly price extracts not committed to the repository. Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the level break, the local slopes, and the three headline kink estimates against the numbers of record before drawing.

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